MIL-H-62430(AT)
26 May 1988
SUPERSEDING
(see 6.5)

MILITARY SPECIFICATION

HARNESS ASSEMBLY, THERMOCOUPLE

This specification is approved for use within the US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Thermocouple Harness Assembly, hereinafter referred to as the harness.

1.2 Classification. The harness shall be of the following types (see 6.2):

Type I - Differentiating characteristics in accordance with Drawing 12302361 and as specified herein.

Type II - Differentiating characteristics in accordance with Drawing 12302362 and as specified herein.

Type III - Differentiating characteristics in accordance with Drawing 12302363 and as specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A
FSC 6685
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:24Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS
FEDERAL

P-C-437	- Cleaning Compound, High Pressure (Steam Cleaner).
VV-F-800	- Fuel Oil, Diesel.

MILITARY

MIL-T-5624	- Turbine Fuel, Aviation, Grades JP-3 and JP-5.
MIL-W-5846	- Wire, Electrical, Chromel and/or Alumel, Thermocouple.
MIL-H-6083	- Hydraulic Fluid, Petroleum Base, for Preservation and Operation.
MIL-L-23699	- Lubricating Oil, Aircraft Turbine Engine, Synthetic Base.
MIL-R-83248	- Rubber, Fluorocarbon Elastomer, High Temperature, Fluid and Compression Set Resistant.

STANDARDS
MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-45662	- Calibration System Requirements.

2.1.2 Other Government documents and drawings. The following documents and drawings form a part of this specification to the extent specified herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:24Z
Check the source to verify that this is the current version before use.

DRAWINGS
ARMY

12284516	- Rear Engine, Sub-Assembly.
12302361	- Harness Assembly, Thermocouple.
12302362	- Harness Assembly, Thermocouple.
12302363	- Harness Assembly, Thermocouple.

NATIONAL BUREAU OF STANDARDS

NBS CIRCULAR 561 - Reference Tables for Thermocouples.

(Application for copies of National Bureau of Standards documents should be addressed to the Superintendent of Documents, US Government Printing Office, Washington DC 20402.)

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. The issue of the document which is indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARD INSTITUTE (ANSI)

ANSI-MC96.1 - Temperature Measurement Thermoucouple.

(Copies of the above publication may be obtained from Instrument Society of America, 67 Alexander Drive, P.O. Box 12277, Research Triangle park, NC 27709.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), the contractor shall furnish harnesses which shall be subjected to first article inspection (see 4.4). First article inspection samples, properly marked with identifying information, shall be representative of the units to be furnished to the Government. All subsequent harnesses delivered to the Government shall conform to these samples in all of their pertinent physical and performance attributes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:24Z
Check the source to verify that this is the current version before use.

3.2 Materials. Materials shall be as specified herein and as noted on the applicable drawing. Materials not designated on the drawing or specifically covered by this specification shall be uniform in quality and free from imperfections or defects which affect their serviceability (see 4.6.1).

3.2.1 Fluid resistance. All materials used in the harness shall be impervious to fluids in accordance with P-C-437, VV-F-800, MIL-T-5624, MIL-H-6083, and MIL-L-23699 (see 4.6.1).

3.2.2 Corrosion resistance. All external surfaces shall be of corrosion-resistant materials or shall be properly protected to prevent corrosion (see 4.6.1).

3.2.3 Thermocouple wire. The thermocouple wire shall be chromel and alumel conforming to the electromotive force requirements of MIL-W-5846 and having an accuracy of plus or minus (±) 0.75 percent (%) from 752 to 1832 degrees Fahrenheit (°F) (400 to 1000 degrees Celsius [°C]) (see 4.6.1).

3.2.4 Conductors and sockets. Chromel and alumel conductors and sockets shall be used throughout the harness (see 4.6.1).

3.2.5 Connector inserts. Connector insert material shall be fluorocarbon in accordance with MIL-R-83248, class I, or equivalent (see 4.6.1).

3.2.6 Thermocouple probes. Thermocouple probes shall be closed and ungrounded with a recovery factor of less than 0.75 (see 4.6.1).

3.2.7 Potting. Flurocarbon in accordance with MIL-R-83248, class I, or equivalent, shall be used as the potting material. Epoxy shall be used in the area where the sockets and conductors are connected to provide additional support (see 4.6.1).

3.2.8 Recycled, virgin, and reclaimed materials. There are no requirements for the exclusive use of virgin materials. The use of recycled or reclaimed (recovered) materials is acceptable provided that all other requirements of this specification are met (see 4.6.1 and 6.3.1).

3.3 Design and construction.

3.3.1 Configuration. The harness shall conform to the applicable drawing (see 4.6.1 and 4.6.2).

3.3.1.1 Schematic arrangemenet. The schematic arrangement for the harness shall be as defined on the applicable drawing (see 4.6.1 and 4.6.2).

3.3.1.2 Interface definition. Electrical and mechanical interfaces shall be as defined on the applicable drawing (see 4.6.1 and 4.6.2).

3.3.2 Weight. The weight of the harness shall not exceed 1.0 pound (see 4.6.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:24Z
Check the source to verify that this is the current version before use.

3.4 Performance.

3.4.1 Individual circuit resistance. The individual circuit resistance between connector contact A and any other contact (except F) at 70°F (21°C), shall be in accordance with the following (see 4.6.4):

a. Type I - 3.19 to 3.67 ohms.
b. Type II - 5.04 to 5.80 ohms.
c. Type III - 5.37 to 6.17 ohms.

3.4.2 Average circuit resistance. The average circuit resistance of the between connector contact A and all other contacts (except F) shorted together at 70°F (21°C) shall be in accordance with the following (see 4.6.5):

a. Type I - 1.92 to 2.28 ohms.
b. Type II - 3.43 to 3.95 ohms.
c. Type III - 3.69 to 4.25 ohms.

3.4.3 Insulation resistance. The insulation resistance between the circuits and connector shells and outer braid shall be not less than 25,000 ohms at 250 volts direct current (V dc) (see 4.6.6).

3.4.4 Output signal. The individual thermocouples shall have an output signal in accordance with Circular 561 (see 4.6.7).

3.4.5 Output signal accuracy. Each individual thermocouple shall produce an output signal within ± 11°F (6°C) of the applied temperature, when the thermocouple junction is subjected to an applied temperature of 1112 ± 18°F (600 ± 10°C) (see 4.6.8).

3.4.6 Leakage. The full length of sensing probe sheath shall be free of flaws or holes that would allow fuel leakage to enter the inner insulation. The leakage rate shall not exceed 3000 cubic centimeters (cc) per minute at 200 ± 10 pounds per square inch (psi) (see 4.6.9).

3.5 Environmental conditions.

3.5.1 High temperature. The harness shall meet the specified performance requirements when exposed to the following temperatures (see 4.6.10.1):

a. Electrical connectors - not greater than 400°F (204°C).
b. Harness and leads - not greater than 1200°F (649°C).
c. Thermocouple probes - 1800 ± 50°F (982 ± 28°C).

3.5.2 Low temperature. The harness shall meet the specified performance requirements when exposed to a temperature of minus 60°F (-54°C) (see 4.6.10.2).

3.5.3 Altitude. The harness shall meet the specified performance requirements when exposed to an altitude of 1300 feet below sea level to 10,000 feet above sea level (see 4.6.10.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:24Z
Check the source to verify that this is the current version before use.

3.5.4 Humidity. The harness shall meet the specified performance requirements, and shall evidence no corrosion or other defects which would affect the function or structural strength, when exposed to zero to 100% relative humidity (see 4.6.10.4).

3.5.5 Salt fog. The harness shall meet the specified performance requirements when tested as specified in 4.6.10.5.

3.5.6 Sand and dust. The harness shall meet the specified performance requirements when tested as specified in 4.6.10.6.

3.5.7 Fungus. The harness shall meet the specified performance requirements when tested as specified in 4.6.10.7.

3.5.8 Vibration. The harness shall meet the specified performance requirements when subjected to vibration having levels in accordance with figure 1 from 25 to 500 to 25 hertz (Hz) (see 4.6.10.8).

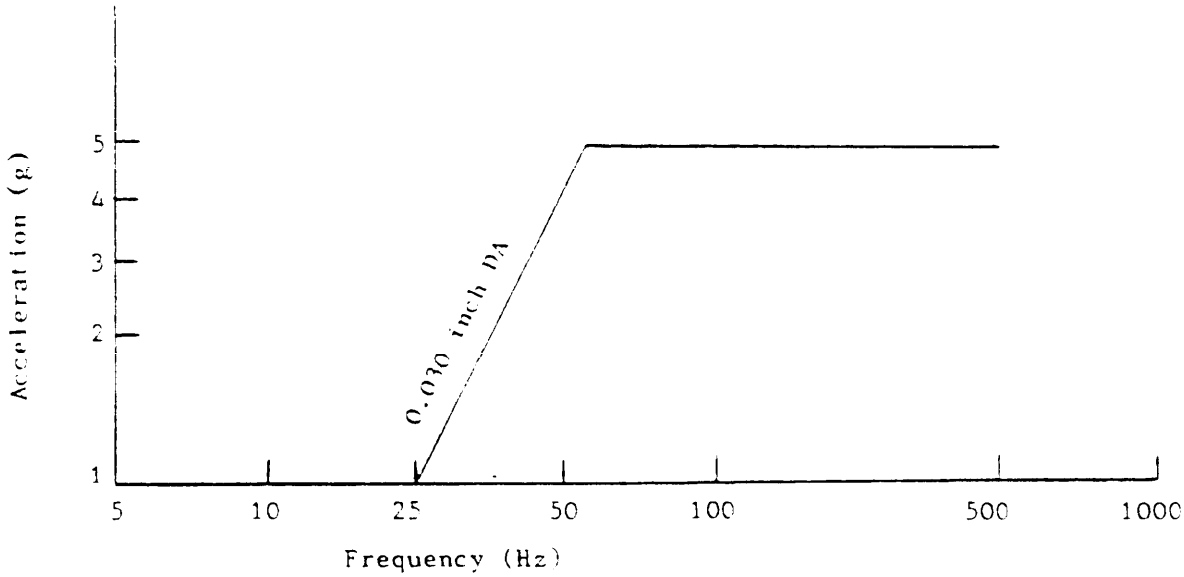


FIGURE 1. Peak Vibration Envelope.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:24Z
Check the source to verify that this is the current version before use.

3.5.9 Gun-firing shock. The harness shall meet the specified performance requirements without malfunction when subjected to shocks of 55 ± 5.5 gravity units (g) for 2.5 ± 0.2 milliseconds (ms) duration in any direction (see 4.6.10.9).

3.5.10 Ballistic shock. The harness shall meet the specified performance requirements without malfunction when subjected to half sine wave shock pulses of 200 ± 20g for 0.5 ± 0.1 ms duration in any direction (see 4.6.10.10).

3.5.11 Basic shock. The harness shall meet the specified performance requirements without malfunction when subjected to half sine wave shock pulses of 30 ± 3g for 18 ms duration in any direction (see 4.6.10.11).

3.5.12 Submergence. The harness shall meet performance requirements after submergence in 8 feet of water (see 4.6.10.12).

3.6 Identification marking. Identification marking shall be in accordance with MIL-STD-130 and as noted on the applicable drawing (see 4.6.2).

3.7 Workmanship. Workmanship shall be in accordance with MIL-STD-454, requirement 9 (see 4.6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection. system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsbility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, eitner indicated or actual, nor does it commit the Government to acceptance of defective material.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:24Z
Check the source to verify that this is the currentversion before use.

4.1.2 Inspection equipment. Unless otherwise specified in the contract (see 6.2), the contractor is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Inspection equipment must be capable of repetitive measurements to an accuracy of 10% of the measurement tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662.

4.2 Inspection conditions. Unless otherwise specivied (see 6.2), all inspections shall be conducted under the following conditions:

a. Temperature: 73 ± 18°F (23 ± 10°C).
b. Pressure: 28.5 (+2, -3) inches (725 [+50, -75] mm) mercury.
c. Relative humidity: 50 ± 30%.

4.2.1 Test tolerances. Unless otherwise specified (see 6.2), test conditions, as indicated by instrumentation, shall not deviate from the conditions specified herein by more than the following:

a. Temperature: Under 100°F (38°C): ± 4°F (2.2°C).
   Over 100°F (38°C): ± 4%.
b. Pressure: ± 5%.
c. Relative humidity: ± 5%.
d. Vibration amplitude:
   (1) Random: ± 3 decibels at resonance and
   ± 1 decibel elsewhere.
   (2) Sinusoidal: ± 2%.
e. Vibration frequency: ± 2%.
f. Shock: ± 5%.
g. Acceleration: ± 5%.
h. Voltage, current, and time:
   (1) Value specified as a whole number: ± 0.5.
   (2) Value specified to 1 decimal place: ± 0.05.
   (3) Value specified to 2 decimal places: ± 0.005.

4.3 Classification of inspection. Inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspections (see 4.5).
   1. Examination (4.5.2).
   2. Acceptance tests (see 4.5.3).

4.4 First article inspection. Unless otherwise specified (see 6.2), the Government shall select three harnesses produced under the production contract for first article inspection. First article samples shall be inspected as specified in table I. Approval of the first article samples by the Government shall not relieve the contractor of his obligation to supply harnesses that are fully representative of those inspected as first article samples. Any changes or deviation of the production units from the first article samples shall be subject to the approval of the contracting officer.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:24Z
Check the source to verify that this is the current version before use.

TABLE I. Classification of inspections.

Title	Requirement	Inspection	First article	Quality conformance	
				Examination	Acceptance (100%)
Materials and construction	3.2 thru 3.3.1.2	4.6.1	X		
Defects (see table II)	3.3.1 thru 3.3.1.2, 3.6 and 3.7	4.6.2	X	X	
Weight	3.3.2	4.6.3	X		
Individual circuit resistance	3.4.1	4.6.4	X		X
Average circuit resistance	3.4.2	4.6.5	X		X
Insulation resistance	3.4.3	4.6.6	X		X
Output signal	3.4.4	4.6.7	X		X
Output signal accuracy	3.4.5	4.6.8	X		X
Leakage	3.4.6	4.6.9	X		X
High temperature	3.5.1	4.6.10.1	X		
Low temperature	3.5.2	4.6.10.2	X		
Altitude	3.5.3	4.6.10.3	X		
Humidity	3.5.4	4.6.10.4	X		
Salt fog	3.5.5	4.6.10.5	X		
Sand and dust	3.5.6	4.6.10.6	X		
Fungus	3.5.7	4.6.10.7	X		
Vibration	3.5.8	4.6.10.8	X		
Gun-firing shock	3.5.9	4.6.10.9	X		
Ballistic shock	3.5.10	4.6.10.10	X		
Basic shock	3.5.11	4.6.10.11	X		
Submergence	3.5.12	4.6.10.12	X		

4.4.1 Failure. Deficiencies found during, or as a result of, first article inspection shall be cause for rejection of the first article samples until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. Any deficiency found during, or as a result of, first article inspection shall be evidence that all items already produced prior to completion of the first article inspection are similarly deficient unless contrary evidence satisfactory to the contracting officer is furnished by the contractor. Such deficiencies on all items shall be corrected by the contractor. The Government will not accept products until first article inspection is completed to the satisfaction of the Government.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:24Z
Check the source to verify that this is the current version before use.

4.5 Quality conformance inspections.

4.5.1 Sampling.

4.5.1.1 Lot formation. An inspection lot shall consist of all the harnesses of one type and part number, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.5.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with general inspection level II of MIL-STD-105.

4.5.2 Examination.

4.5.2.1 Acceptable quality level. Each sample selected in accordance with 4.5.1.2 shall be examined to determine conformance to the following acceptable quality levels (AQL) on the basis of % defective:

Classification	AQL
Major	1.0
Minor	2.5

4.5.2.2 Classification of defects. For examination purposes, defects shall be classified as listed in table II.

TABLE II. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major	AQL 1.0% Defective	
101	Assembly, incomplete (see 3.3.1).	Visual
102	Dimensions affecting interchangeability, out of tolerance (see 3.3.1).	SIE 1/
103	Workmanship, faulty affecting performance (see 3.7).	Visual
Minor	AQL 2.5% Defective	
201	Dimensions not affecting interchangeability, out of tolerance (see 3.3.1).	SIE
202	Marking, improper (see 3.6).	Visual
203	Workmanship, faulty affecting appearance (see 3.7).	Visual

1/ SIE = Standard Inspection Equipment.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:24Z
Check the source to verify that this is the currentversion before use.

4.5.2.3 Failure. If an inspection lot is rejected, the contractor may repair the defects, or screen out the defective items, and submit the lot for reinspection. Such lots shall be inspected using tightened inspection, shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.5.3 Acceptance tests (100%). Each harness shall be subjected to the acceptance tests specified in table I.

4.5.3.1 Failure. Failure of any harness to pass any of the acceptance tests shall be cause for rejection of the failed item unless authorization is obtained from the Government to repair the failed item. All repaired items shall be subjected to the acceptance tests specified in table I.

4.6 Methods of inspection

4.6.1 Materials and construction. Conformance to 3.2 thru 3.3.1.2 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.6.2 Defects. Conformance to 3.3.1 thru 3.3.1.2, 3.6, and 3.7 shall be determined by examination for the defects listed in table II. Examination shall be visual, tactile, or by measurement with SIE.

4.6.3 Weight. To determine conformance to 3.3.2, the harness shall be weighed.

4.6.4 Individual circuit resistance. To determine conformance to 3.4.1, the individual circuit resistance shall be measured.

4.6.5 Average circuit resistance. To determine conformance to 3.4.2, the average circuit resistance shall be measured.

4.6.6 Insulation resistance. To determine conformance to 3.4.3, the insulation resistance between the individual circuits and connector shells and outer braid shall be measured at an applied potential of 250 V dc.

4.6.7 Output signal. To determine conformance to 3.4.4, the output signal of the individual thermocouples shall be measured under the conditions specified in Circular 561.

4.6.8 Output signal accuracy. To determine conformance to 3.4.5, each individual thermocouple junction shall be subjected to an applied temperature of 1112 $\pm$ 18°F (600 $\pm$ 10°C) and the output signal measured.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:24Z
Check the source to verify that this is the current version before use.

4.6.9 Leakage. To determine conformance to 3.4.6, a leak test shall be performed under the conditions specified in 3.4.6.

4.6.10 Environmental tests. The harness shall be subjected to the environmental tests specified herein. After completion of each environmental test, the performance tests specified in 4.6.4 thru 4.6.9 shall be repeated. Prior to conducting the environmental tests, the harness shall be subjected to an accelerated aging temperature of 392°F (200°C) for 168 hours.

4.6.10.1 High temperature. To determine conformance to 3.5.1a and 3.5.1b, the harness shall be subjected to the high-temperature test of MIL-STD-810, method 501.1, procedure I, except steps 2 and 3 shall be ommited and the temperatures specified in 3.5.1 shall apply. The harness shall be maintained at the temperatures for an 8-hour period.

4.6.10.1.1 Thermocouple probe cycling. To determine conformance to 3.5.1c, upon completion of the high-temperature test, the thermocouple probes shall be subjected to not less than 8000 cycles from a room temperature of 77°F (25°C) to 1800 ± 50°F (982 ± 28°C).

4.6.10.2 Low temperature. To determine conformance to 3.5.2, the harness shall be subjected to the low-temperature test of MIL-STD-810, method 502.1, procedure I, except steps 2 and 3 shall be omitted. The harness shall be placed dry in a temperature chamber with the ambient temperature maintained at -60 ± 10°F (-54 ± 6°C) for 4 hours.

4.6.10.3 Altitude range. To determine conformance to 3.5.3, the harness shall be subjected to an atmospheric pressure of 31.19 inches of mercury (1300 feet below sea level), after which the harness shall be subjected to an atmospheric pressure of 20.01 inches of mercury (10,000 feet above sea level).

4.6.10.4 Humidity. To determine conformance to 3.5.4, the harness shall be subjected to the humidity test of MIL-STD-810, method 507.1, procedure II. At the conclusion of the humidity test, the harness shall be dried without forced convection for 12 hours in air at less than 70°F (21°C) with not less than 50% relative humidity and then subjected to the performance tests. After this, the harness shall be disassembled and inspected.

4.6.10.5 Salt fog. To determine conformance to 3.5.5, the harness shall be subjected to the salt fog test of MIL-STD-810, method 509.1, procedure I.

4.6.10.6 Sand and dust. To determine conformance to 3.5.5, the harness shall be subjected to the sand and dust test of MIL-STD-810, method 510.1, procedure I.

4.6.10.7 Fungus. To determine conformance to 3.5.7, the harness shall be tested in accordance with MIL-STD-810, method 508.1, procedure I. In lieu of the performance of the MIL-STD-810 test, certification of compliance, with supporting data, may be provided attesting that the harness is constructed of materials that would not support fungus growth.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:24Z
Check the source to verify that this is the current version before use.

4.6.10.8 Vibration. To determine conformance to 3.5.8, the harness shall be mounted using the installed interface configuration for its intended application, or equivalent, and subjected to the test of MIL-STD-810, method 514.2, procedure VIII, except that the conditions of 3.5.8 shall apply. A 20-minute logarithmic resonant sweep shall be conducted in each of three mutually perpendicular axes, from 25 to 500 to 25 Hz, to identify up to four of the most severe resonant frequencies. A resonant dwell shall then be conducted for 20 minutes at each of the four resonant frequencies, for a total of 100 minutes in each axis. If less than four resonant frequencies are encountered, the balance of the 100 minutes vibration time in each axis shall be spent cycling the harness over the specified frequency range.

4.6.10.9 Gun-firing shock. To determine conformance to 3.5.9, the harness shall be mounted using the installed interface configuration for its intended application, or equivalent, and subjected to the test of MIL-STD-810, method 516.2, procedure IV, except that conditions of 3.5.9 shall apply. A total of three half sine wave shock pulses shall be applied in both directions along each of three mutually perpendicular axes. Peak amplitude shall be 55 $\pm$ 5.5g for 2.5 $\pm$ 0.2 ms.

4.6.10.10 Ballistic shock. To determine conformance to 3.5.10, the harness shall be tested in accordance with MIL-STD-810, method 516.2, procedure IV, except that conditions of 3.5.10 shall apply. The harness shall be mounted normally in a nonoperational mode and subjected to three half sine wave shocks of 200 $\pm$ 20g for 0.5 $\pm$ 0.1 ms in each direction along each of three mutually perpendicular axes.

4.6.10.11 Basic shock. To determine conformance to 3.5.11, the harness shall be mounted using the installed interface configuration for its intended application, or equivalent, and subjected to the test of MIL-STD-810, method 516.2, procedure IV, except that conditions of 3.5.11 shall apply. A total of three half sine wave shock pulses shall be applied in both directions along each of three mutually perpendicular axes. Peak amplitude shall be 30 $\pm$ 3g for 18 ms.

4.6.10.12 Submergence. To determine conformance to 3.5.12, all electrical fittings shall be capped and the harness shall be subjected to a 1-hour cycle test consisting of immersion in 70°F water at a depth of 6 feet to 8 feet at a cycle rate of 4 cycles per minutes. Within 2 hours after completion, the harness shall be subjected to the specified performance tests.

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level shall be in accordance with the applicable packaging requirements specified by the contracting authority (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:24Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use. The harness covered by this specification is intended to be used to generate signals required for measurement of gas temperature at the power turbine stator entry on the AGT 1500 gas-turbine engine.

6.1.1 Item definition. The harness consists of an integral four-probe harness, lead, and output connector assembly. Individual readout signal for each thermocouple is available at the output connector. The individual circuit electrical resistance need not be the same.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, revision, and date of this specification.
b. Title, number, revision, and date of applicable drawings.
c. Type of harness required (see 1.2).
d. If first article inspection is not required (see 3.1).
e. If responsibility for inspection shall be other than as specified (see 4.1).
f. If responsibility for inspection equipment shall be other than as specified (see 4.1.2).
g. If inspection conditions shall be other than as specified (see 4.2).
h. If test tolerances shall be other than as specified (see 4.2.1).
i. If first article sample size shall be other than as specified (see 4.4).
j. Selection of applicable level and packaging requirements (see 5.1).

6.3 Definitions.

6.3.1 Recovered materials. "Recovered materials" means materials that have been collected or recovered from solid waste (see 6.3.2).

6.3.2 Solid waste. "Solid waste" means (a) any garbage, refuse, or sludge from a waste treatment plant, water supply treatment plant, or air pollution control facility; and (b) other discarded material, including solid, liquid, semisolid, or contained gaseous material resulting from industrial, commercial, mining, and agricultural operations, and from community activities. It does not include solid or dissolved material in domestic sewage, or solid or dissolved material in irrigation return flows or industrial discharges which are point sources subject to permits under section 402 of the Clean Water Act, (33 U.S.C. 1342 et seq.), or source, special nuclear, or byproduct material as defined by the Atomic Energy Act of 1954 (42 U.S.C. 2011 et seq.) (Source: Federal Acquisition Regulations, section 23.402).

6.4 Subject term (key word) listing.

Harness
Thermocouple.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:24Z
Check the source to verify that this is the current version before use.

6.5 Superssion data. This specification supersedes AVCO Lycoming specification E2137B, dated 26 April 1982.

Custodian:
Army - AT

Preparing activity:
Army - AT

(Project 6685-A821)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:24Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300
AMSTA-GDS

Commander
US Army Tank-Automotive Command
ATTN: AMSTA-GDS
Warren, MI 48397-5000

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:24Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS This form is provided to solicit beneficial comments which may improve this document and enhance its use DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

FSC 6685

DOCUMENT IDENTIFIER (Number) AND TITLE

MIL-H-62430(AT); Harness Assembly, Thermocouple

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1 ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID RESTRICTIVE LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDED WORDING CHANGE

C REASON FOR RECOMMENDED CHANGE(S)

2 REMARKS

SUBMITTED BY (Printed or typed name and address — Optional)

TELEPHONE NO

DATE

DD FORM 1 OCT 76 1426

EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:24Z
Check the source to verify that this is the current version before use.

an irritant to the eyes and respiratory tract and can be toxic as